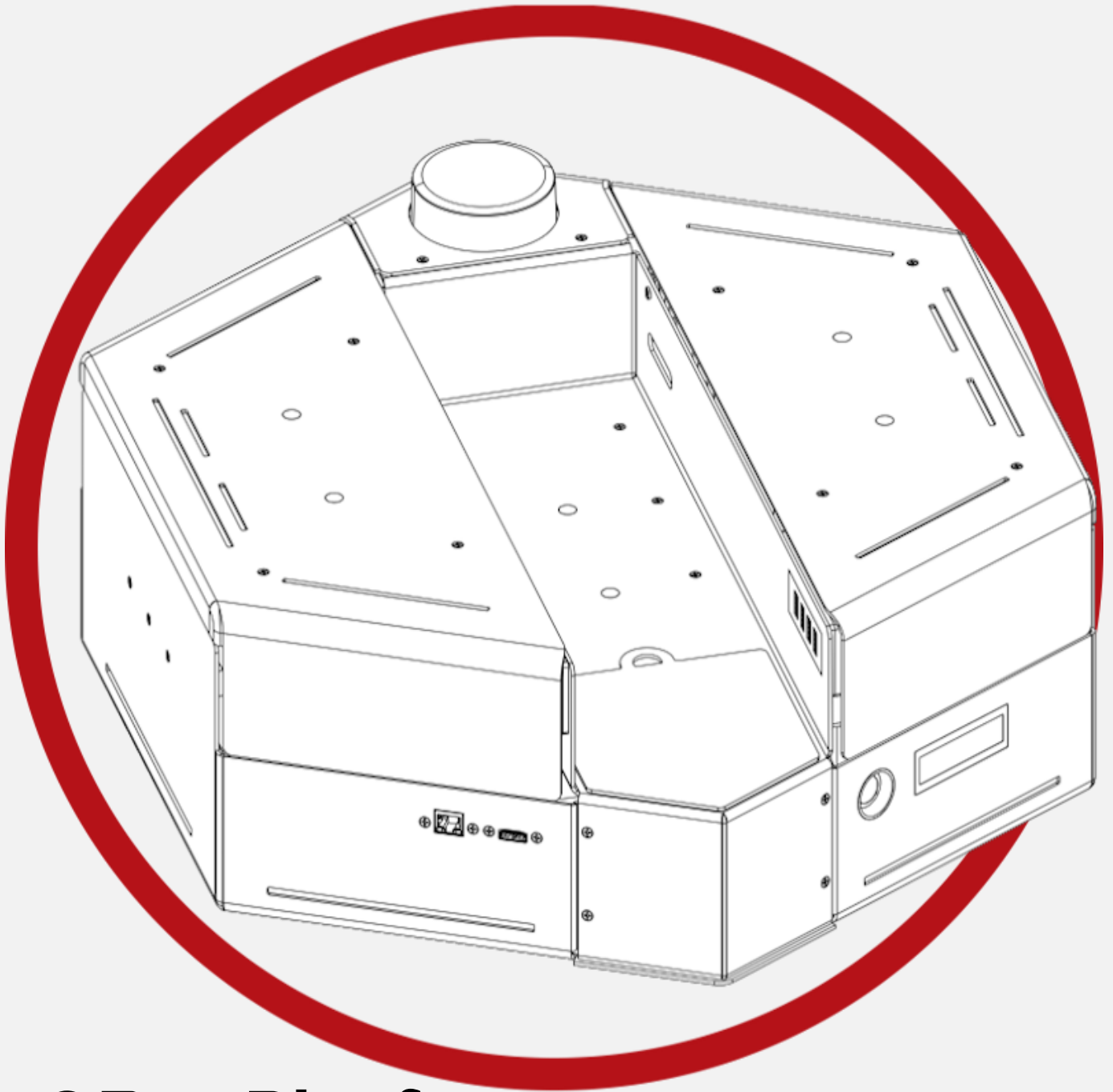




QBot Platform

Sensor Fusion

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QBot Platform – Application Guide

Sensor Fusion

Why Explore Sensor Fusion?

Sensor fusion allows us to combine data from multiple sensors to achieve a more accurate and complete estimate than each sensor could provide individually. These techniques allow us to rely on the strengths of each sensor to compensate for the limitations of others. This makes sensor fusion useful in applications such as localization of mobile robots, traffic detection in autonomous vehicles, and health monitoring in wearable consumer electronics.

Sensor Fusion

This is the second lab of Skills Progression 2: Surveying. In this Skills Progression, the line following and obstacle detection functionality developed during Skills Progression 1 is excluded from the code for simplicity. You will use the same joystick or keyboard controls from the previous Skills Progressions to control the motion of the QBot Platform.

In this lab, you will use sensor fusion techniques to perform state estimation of the QBot Platform fusing data from multiple sensors using a complementary filter. This lab builds on Lab 1: Localization, where you estimated the position of the QBot by running Scan Match on LiDAR data.

Note: code for this skills progression does not include implementations for obstacle avoidance. You are required to ensure the QBot does not run into obstacles while manually driving it.

Before you begin

Please review the following before beginning this lab,

1. Ensure you have completed the following labs in your language of choice:
 - a. **Play** lab from Skills Progressions 0
 - b. **Obstacle Detection** lab from Skills Progression 1
 - c. **Localization** lab from Skills Progression 2
2. Ensure that you have read the **Sensor Fusion** concept review.
3. If you are completing the lab with hardware, set up an environment by arranging walls surrounding the QBot with some unique features. A sample wall layout is shown in Figure 1:

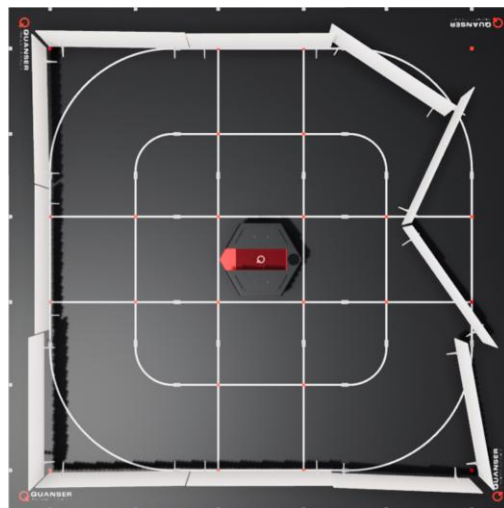


Figure 1: Sample wall layout